

DEPARTMENT OF ECONOMICS

MACROECONOMICS (ECN 1215) APRIL 2017

TUTORIAL SHEET 4 SOLUTIONS

1. Suppose in a simple economy with no foreign sector, the mpc equals 0.8. intended investment spending has suddenly fallen, reducing AD and output to a level that is 100 million below Y^*
- a) If the government decide to try to get the economy back to full employment using only an increase in government spending (ΔG), by how much would G need to be increased?
 - b) If the government, instead, decided to try to get the economy to full employment using only a lump-sum tax cut (ΔT), how big of a tax cut would be needed?
 - c) Alternatively, if the government decided to try to get the economy back to full employment using only an increase in transfers (ΔTR), how large would this increase need to be?

a) $multiplier = \frac{1}{mps} = \frac{1}{0.2} = 5$

$$\Delta Y = \Delta G \times 5$$

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$$100m = 5 \times \Delta G$$

$$\Delta G = 20m$$

b) $Tax\ multiplier = -mpc \times spending\ multiplier$

$$= -0.8 \times 5$$

$$= -4$$

$$\Delta Y = \Delta T \times (-mpc.m)$$

$$100 = \Delta T \times -4$$

$$\Delta T = \frac{100m}{-4}$$

$$= -25m$$

Therefore, output reduces by 25m

c)

$$\Delta Y = \Delta TR \times (mpc \times m)$$

$$100m = \Delta TR \times (0.8 \times 5)$$

$$\Delta TR = 25m$$

2. Assume a simple closed economy, with an mpc equal to 0.75. The government has passed a balanced budget amendment. The economy goes into a recession, so the government increases government spending by 40 million to try to expand the economy
- Calculate the change in output (ΔY) from the increase in government spending (ΔG)
 - The balanced budget amendment requires the government to also raise taxes by 40 million. Calculate the change in output (ΔY) from the tax hike
 - What is the net effect on output from these two policies? Was there any expansionary effect?

$$a) \text{ multiplier} = \frac{1}{mps} = \frac{1}{0.25} = 4$$

$$\Delta Y = \Delta G \times m$$

$$= 40 \times 4$$

$$= 160$$

$$b) \text{ Tax multiplier} = -mpc \times \text{spending multiplier} = -0.75 \times 4 = -3$$

$$\Delta Y = \Delta T \times \text{Tax. mult}$$

$$= 40m \times -3$$

$$= -120m$$

Therefore, output reduces by 120m

c) *Net Effect*

$$\Delta Y = 160m + (-120)$$

$$= 160m - 120m$$

$$= 40m$$

Yes, it expanded output by 40m

3. Explain retrogressive, proportional and progressive taxes. Which one(s) is used in Zambia?

Retrogressive tax: *A tax that takes a larger percentage of income from low-income groups than from high-income groups. In other words, a tax whose burden reduces as income increases. Excise duties on smoking and alcohol, or on any good that is mostly consumed by low income populations.*

Proportional tax: *A tax that takes the same percentage or fixed proportion of income from all income groups. E.g corporate tax*

Progressive tax: *A tax where the percentage increases with income. In other words, a tax that takes a larger percentage of income from high-income groups than from low-income groups. E.g income tax (PAYE in Zambia)*

4. Explain the principles for a good tax system?

- *Revenue adequacy*
- *Stability*
- *Simplicity*
- *Tax neutrality*
- *Low administration and compliance cost*
- *Low administration and compliance cost*

(Briefly expand your points in a sentence or two, don't just list)

5. What is fiscal policy? Differentiate between contractionary and expansionary fiscal policies? How can the government use fiscal policy to reduce unemployment?

Fiscal policy *is the use of taxes and government spending, and sometimes government transfers, to shift the aggregate demand curve in order to influence the nation's economy. The two main instruments of fiscal policy are government expenditures and taxes.*

Expansionary fiscal policy *is an increase in government spending and/or a decrease in taxes to increase aggregate demand in a recession or depression. This is usually used to reduce unemployment.*

Contractionary fiscal policy (also referred to as austerity measures or deficit-reduction policies) is defined as a decrease in government spending and/or an increase in taxes to decrease aggregate demand. This is usually used to solve the problem of inflation.

Expansionary fiscal policies make a budget surplus smaller or a budget deficit bigger.

Contractionary fiscal policies make a budget surplus bigger or a budget deficit smaller (this is why it's called an austerity measure). [Note: give examples in a test]

Discretionary fiscal policy: government takes deliberate actions through legislation to alter spending or taxation. [The key word is "deliberate"]

6. Explain the significance of the Laffer curve?

The Laffer curve depicts the relationship between the tax rate and tax revenue. It enables the government to maximize revenue collection by charging the optimal level of taxes without causing people to lose incentives to work hard. At a zero tax rate the tax revenue is zero, but as the tax rate increase tax revenue increases to a point where it starts to drop. Laffer argued that the more money taken from a business in the form of taxes, the less money it has to invest in the business. Business owners will find ways to protect their profits from taxation, or to relocate their operations overseas. If employees see increasing amounts of their income taken due to increased efforts on their part, they will lose the incentive to work harder. For every type of tax, there is a threshold rate above which the incentive to produce more diminishes, thereby reducing the amount of revenue the government receives. [Include diagram from your handout].

7. What are the functions of money?

- 1) Medium of exchange
- 2) Unit of account
- 3) Store of wealth/value

[In a test you briefly explain using your handout notes and provide an example for each]

8. Suppose you are advising a small country on whether to print its own money or to use the money of its larger neighbor (such as the United States). What are the costs and benefits

of a national currency? Does the relative political stability of the two countries have any role in this decision?

The major benefit of having a national currency is seigniorage—the ability of the government to raise revenue by printing money. The major cost is the possibility of inflation, if the government relies too heavily on seigniorage. The benefits and costs of using a foreign currency are exactly the reverse: the benefit of foreign money is that inflation is no longer under domestic political control, but the cost is that the domestic government loses its ability to raise revenue through seigniorage. The foreign country's political stability is a key factor. The primary reason for using another nation's money is to gain stability. If the foreign country is unstable, then the home country is better off using its own currency—the home economy remains more stable, and it keeps the seigniorage.

9. Explain the roles/functions of the central bank and commercial banks in a country?

Central Bank

Conducts monetary policy in order to influence the level of economic activity in the country, and this involves regulating the supply of money and the cost and availability of credit. Other functions include:

- 1) Lender of last resort (comes to the aid of commercial banks when faced with bankruptcy to prevent liquidation.*
- 2) Acting as the banker's bank (regulating commercial banks)*
- 3) The government's bank (keeps government money)*
- 4) Manager of public debt (e.g. selling and buying of treasury bills)*

Commercial Banks

The primary purpose of commercial banks is accepting deposits and lending money. Secondary functions include: issuing letters of credit, travellers cheques; undertaking safe custody of valuables and important documents; transferring money from one place to another; providing customers with facilities of foreign exchange; etc.

10. What is monetary policy?

This is government policy conducted by the central bank so as to influence money supply and the level of economic activity in the economy in order to control inflation and stabilize output.

11. Illustrate how the central bank uses monetary policy tools (open market operations, reserve requirements and the discount rate) to influence money supply and the level of economic activity in the country?

1) Open Market Operations (OMO)

Involves the central bank buying or selling government bonds in the open market.

Expansionary OMO: If the central bank wants to increase money supply and reduce the cost of borrowing (interest rate), it will buy bonds.

Restrictive OMO: If the central bank wants to reduce money supply and increase the cost of borrowing (interest rate), it will sell bonds.

2) Reserve Requirements

Refers to the asset/money balances that commercial banks should keep in reserve, usually 10% of their total bank deposits.

If the central bank wants to increase the money supply and reduce the cost of borrowing (interest rate), it will reduce the reserve requirement ratio.

If the central bank wants to reduce the money supply and increase the cost of borrowing (interest rate), it will increase the reserve requirement ratio.

3) Discount Rate

Refers to the interest rate at which the central bank provides loans to commercial banks.

If the central bank wants to increase money supply and reduce the cost of borrowing (interest rate), it will reduce the discount rate.

If the central bank wants to reduce money supply and increase the cost of borrowing (interest rate), it will increase the discount rate.

[Note: if you're asked to explain how the above 3 tools can be used to control inflation, bring out the points and examples that show how the 3 tools reduce money supply consequently reducing inflation]

12. Explain the difference between expansionary and contractionary monetary policies?

1. Expansionary monetary policy is monetary policy aimed at increasing economic activity in the economy achieved through buying bonds, reducing reserve requirement ratio, or lowering the discount rate. This is aimed at increasing money supply so that people and businesses have more money to invest in order to reduce unemployment [Add an example in a test].
2. Contractionary monetary policy is monetary policy aimed at reducing economic activity achieved through selling bonds, increasing reserved requirement ratio or rising discount rate. This is aimed at reducing money supply in order to slow down inflation [Add an example in a test].

13. Explain the liquidity preference theory? According to this theory what are the motives for holding money?

Liquidity Preference Theory

[Cash money is referred to as liquidity, and people's liking for cash money is called liquidity preference].

Propounded by John Maynard Keynes, the Liquidity Preference Theory states that the interest rate adjusts to balance the supply and demand for money. Interest rates will decrease as people will hold excess cash balances and vice versa.

According to Keynes people demand liquidity or prefer liquidity because they have three different motives for holding cash rather than bonds etc.

1. Transaction Motive

Money held for the purpose of making everyday market purchases. This motive depends on the size of income, time gap between the receipts of income, and spending habits. Demand is inelastic.

2. Precaution Motive

Money held for unexpected market transactions or for emergencies. This motive depends on the size of the income, nature of the person, and foresight.

3. Speculative Motive

Money held for speculative purposes or sudden financial opportunities such as a drop in the prices of bonds or securities.

14. Explain the factors that influence the demand for money?

[The supply of money differs from the demand for money. No private individual can change it. Money supply is controlled by the central bank or its government].

Factors that influence money demand are:

- 1. Price level*
- 2. Interest rate*
- 3. Income*
- 4. Credit availability*

[Expand the points using your notes]

15. Explain how equilibrium interest rates are affected by money demand and supply?

Equilibrium interest rates will rise if there is a rise in money demand but a decrease in money supply. Equilibrium interest rates drop if there is a drop in money demand but a rise in money supply. [Insert diagram from your handout]

16. What is the liquidity trap?

The liquidity trap refers to a state in which the nominal interest rate is close or equal to zero and the central bank is unable to stimulate the economy with monetary policy. When the nominal interest rate is very close or equal to zero, the opportunity cost of holding money becomes zero, and economic agents—banks, firms, or individuals—tend to hoard money even if they have more money than they need for transaction purposes. As a result monetary policy becomes ineffective in stimulating the economy because the money creation process does not function as theory predicts since excess money is not converted into new investment. The end result is recession and deflation. [Include diagram from handout].